MA378 – Numerical Analysis II

Lab 3: Gaussian Quadrature

Goal: Derive Gaussian Quadrature rules and implement them using Matlab

Quadrature Rules are methods of approximating definite integrals, e.g., $\int_a^b f(x)dx$. The $(n+1)$-point Gaussian rule is of the form

$$G_n(f) := w_0 f(x_0) + w_1 f(q_1) + \cdots + w_n f(x_n).$$

where the quadrature points, x_i, and the weights, w_i, are chosen is some "optimal" way.

The approach we take to this is to choose the x_i and the w_i so that rule has precision $2n+1$. This can be done by the *method of undetermined coefficients*, but requires us to solve a system of nonlinear equations. For larger n, computational methods are employed. Here, for small n we can work out the rules by hand.

1. Use the method of undetermined coefficients to derive the rule

$$G_1(f) = w_0 f(x_0) + w_1 f(x_1) \approx \int_{-1}^{1} f(x)dx.$$

   You may refer to your notes and to online sources if required.

2. Try adapting this to derive the 3-point rule $G_2(f)$.

3. Download the Matlab file `NewtonCotes.m` from Blackboard. This implements the 2, 3 and 4-point Newton-Cotes methods to estimate $\int_{-1}^{1} \ln(2+x)dx$. Verify that you get the same results that are given in the table below.

4. Write a similar program to implement the Gaussian rules $G_1(\cdot)$ and $G_2(\cdot)$. Compare their accuracy with the corresponding Newton-Cotes Rule.

   Try implementing $Q_4(\cdot)$ (Newton-Cotes: known as Boole's rule) and $G_3(\cdot)$ rules. Over the interval $[-1,1]$ these are (approximately) as follows:

$$Q_4(f) = \frac{2}{90}(7f_0 + 32f_1 + 12f_2 + 32f_3 + 7f_4).$$

$$G_3(f) = 0.3478f(-0.8611) + 0.6521f(-0.3400) + 0.6521f(0.3400) + 0.3478f(0.8611).$$

Method		Quadrature Points (x_0, x_1, etc)	Weights (w_0, w_1, etc)	Estimate	Error
Newton-Cotes	$Q_1(\cdot)$	$-1, 1$	$1, 1$	1.098612	1.972×10^{-1}
Newton-Cotes	$Q_2(\cdot)$	$-1, 0, 1,$	$\frac{1}{3}, \frac{4}{3}, \frac{1}{3}$	1.290400	5.437×10^{-3}
Newton-Cotes	$Q_3(\cdot)$	$-1, -\frac{1}{3}, \frac{1}{3}, 1$	$\frac{1}{4}, \frac{3}{4}, \frac{3}{4}, \frac{1}{4}$	1.293246	2.591×10^{-3}
Newton-Cotes	$Q_4(\cdot)$				
Gaussian	$G_1(\cdot)$				
Gaussian	$G_2(\cdot)$				
Gaussian	$G_3(\cdot)$				

When you are done, upload your Matlab code that has been edited to implement G_1, G_2, G_3 and Q_4 to the relevant Assessment on Blackboard.

Deadline: 5pm, Friday, 7 May.